§ Calogero-Moser system (rational).

- Want to study $M = T^*(\mathbb{C}^n \setminus \Delta) = \{(p_1,\dots,p_n,q_1,\dots,q_n) \mid q_i \neq q_j\}$

With Poisson bracket $\{p_i,q_j\}=\delta_{ij}$, $\{p_i,p_j\}=\{q_i,q_j\}=0 \ \forall i,j$.

And Hamiltonian: $H = \sum_{i=1}^n \frac{p_i^2}{2} + \sum_{i\neq j} \frac{g^2}{(q_i-q_j)^2}$ $(g\in\mathbb{C})$

- Goal: to include H to an integrable system.

- Lax matrix $L_{ij} = \delta_{ij} p_i + ig\frac{(1-\delta_{ij})}{q_i-q_j} = \begin{pmatrix} p_1 & \frac{ig}{q_1-q_2} & \cdots & \frac{ig}{q_1-q_n} \\ \frac{ig}{q_2-q_1} & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & \frac{ig}{q_{n-1}-q_n} \\ \frac{ig}{q_n-q_1} & \cdots & \frac{ig}{q_n-q_{n-1}} & p_n \end{pmatrix}_{ij}$

$$L = \sum_{i=1}^n p_i E_{ii} + ig\sum_{i\neq j} \frac{E_{ij}}{q_i-q_j}.$$

- r-matrix

$$r_{12} = \sum_{i\neq j} \frac{1}{q_i-q_j} E_{ij}\otimes E_{ji} + \sum_{i\neq j} \frac{1}{q_i-q_j} E_{ii}\otimes E_{ij}.$$

Write $L = p + \underbrace{\tilde{L}}_{r\text{ diag. part}}$

Lemma: $\{L_1, L_2\} = [r_{12}, L_1] - [r_{21}, L_2]$.

Pf. Computation. (Exercise) ∎

Theorem Let $H_k := \frac{1}{k}\mathrm{Tr}(L^k)$. Then

1) $H_2 = H$

2) $\{H_1,\dots,H_n\}$ is an integrable system.

Proof:

Remains to prove functional independence.

Note that $H_k = \frac{1}{k}\mathrm{Tr}\left((p+\tilde{L})^k\right) = \frac{1}{k}\sum_{i=1}^n p_i^k + \text{lower terms in } p_1,\dots,p_n$.

$$\Rightarrow \partial_{p_i}(H_k) = p_i^{k-1} + \text{lower terms in } p_1,\dots,p_n$$

$$\begin{pmatrix} dH_1 \\ \vdots \\ dH_n \end{pmatrix} = \begin{pmatrix} 1 & \cdots & 1 \\ p_1 & \cdots & p_n \\ p_1^2 & \cdots & p_n^2 \\ \vdots & & \vdots \\ p_1^{n-1} & \cdots & p_n^{n-1} \end{pmatrix} \cdot \begin{pmatrix} \ast & & \\ & \ddots & \\ & & \ast \end{pmatrix}$$

$$\Rightarrow \begin{vmatrix} 1 & & 1 \\ p_1+\cdots & \cdots & p_n+\cdots \\ \vdots & & \vdots \\ \cdots & \cdots & \cdots \end{vmatrix} = \begin{vmatrix} 1 & \cdots & 1 \\ p_1 & & p_n \\ p_1^2 & & \ddots \\ \vdots & & \ddots \\ p_1^{n-1} & & p_n^{n-1} \end{vmatrix} + \text{lower terms} = \prod_{i>j}(p_i - p_j) + \underset{\text{terms}}{\text{lower}} \neq 0$$

$\Rightarrow$ At generic point $\operatorname{rk}\begin{pmatrix} dH_1 \\ \vdots \\ dH_n \end{pmatrix} = n$

∎

§2 Recall on Gaudin model

We study $\mathcal{B}^{\nu}$ acting on $\bigotimes_{i=1}^{n} L_{w_i}(q_i)$, $\nu = \begin{pmatrix} \nu_1 & & 0 \\ & \ddots & \\ 0 & & \nu_N \end{pmatrix}$

Bethe algebra:

$$\operatorname{cdet}\left(\partial - \nu - E(x)\right) = \partial^N + B_1(x)\partial^{N-1} + B_2(x)\partial^{N-2} + \dots .$$

$\left(E(x)\right)_{ij} = E_{ij}(x)$. Bethe algebra is commutative.

One can show (Exercise)

$$B_2(x) = -\frac{1}{2}\sum_{a,b=1}^{N} E_{ab}(x)E_{ba}(x) - \sum_{a=1}^{N} \nu_a E_{aa}(x) + \text{center} + (\dots)'(x).$$

Then

$$H_i^G = \operatorname*{Res}_{x=q_i}\left(-B_2(x)\right) = \sum_{a,b=1}^{N}\sum_{\substack{k\neq i \\ k=1}}^{n} \frac{E_{ab}^{(i)} E_{ba}^{(k)}}{q_i - q_k} + \sum_{a=1}^{N} \nu_a E_{aa}^{(i)} .$$

- Gaudin Hamiltonians.

Bethe ansatz:

Prepare $N-1$ groups of variables (aka colors)

$$t_1^{(1)}, \dots, t_1^{(N_1)} \,\Big|\, t_2^{(1)}, \dots, t_2^{(N_2)} \,\Big|\, \dots \,\Big|\, t_{N-1}^{(1)}, \dots, t_{N-1}^{(N_{N-1})} .$$

Master function:

$$\Phi(t,q) = e^{-\sum_{a=1}^{N}\sum_{\beta}(\nu_a - \nu_{a+1})t_a^{(\beta)}} \cdot \prod_{\substack{\beta\neq\beta' \\ a}} \left(t_a^{(\beta)} - t_a^{(\beta')}\right)^2 \prod_{\substack{a,\beta, \\ \beta'}} \left(t_a^{(\beta)} - t_{a-1}^{(\beta')}\right)^{-1} \cdot$$

$$\cdot \prod_{a,\beta,\beta'} \left(t_a^{(\beta)} - t_{a-1}^{(\beta')}\right)^{-1} \prod_{\beta,i} \left(t_1^{(\beta)} - q_i\right)^{-1} .$$

BAE

$$v_a - v_{a+1} - 2\sum_{\beta' \neq \beta} \frac{1}{t_a^{(\beta)} - t_a^{(\beta')}} + \sum_{\beta'} \frac{1}{t_a^{(\beta)} - t_{a+1}^{(\beta')}} + \sum_{\beta'} \frac{1}{t_a^{(\beta)} - t_{a-1}^{(\beta')}} -$$

$$- \delta_{a,1} \cdot \sum_{i=1}^{n} \frac{1}{t_a^{(\beta)} - q_i} = 0 \,, \quad \forall a, \beta.$$

Spectrum Gaudin Hamiltonians:

$$H_i \longmapsto v_1 + \sum_{k : k \neq i} \frac{1}{q_i - q_k} - \sum_{a=1}^{M} \frac{1}{q_i - t_1^{(a)}} =: h_i$$

Goal:

Main theorem. (Quantum-classical duality)

Let $\{p_i\}$, $\{q_i\}$

$$L_1(\{p_1,\dots,p_n\},\{q_1,\dots,q_n\}) = \delta_{ij}\, p_i + \frac{(1-\delta_{ij})}{q_i - q_j}$$

Let $(t_a^{(\beta)})$ be a sol-n of BAE:

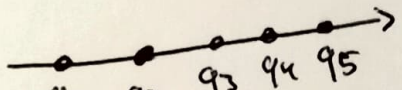
Then

$$\mathrm{Spec}\left(L_1\left(\{H_i((t_a^{(\beta)}))\},\{q_i\}\right)\right) = \{\underbrace{v_1,\dots,v_1}_{n-N_1},\underbrace{v_2,\dots,v_2}_{N_1-N_2},\dots,\underbrace{v_N,\dots,v_N}_{N_{N-1}}\}$$

Cor Spectrum can be found by solving inverse spectral problem (of Gaudin) for L_1:

Cor. If $(q_1(t),\dots,q_n(t))$ is evolution of CM particles, then Spectrum of Gaudin model evolves as $(p_1(t),\dots,p_n(t)) = (\dot{q}_1(t),\dots,\dot{q}_n(t))$

"Picture".



Proposition

Fix two sets of variables: $\{q_i\}_{i=1}^{n} = q$, $\{t_a\}_{a=1}^{M} = t$

Let

$$\mathcal{L}_{ij} = \delta_{ij}\left(\omega + \sum_{k:k\neq i}\frac{1}{q_i - q_k} + \sum_{a=1}^{M}\frac{1}{t_a - q_i}\right) + (1-\delta_{ij})\frac{1}{q_i - q_j} \quad (n\times n)$$

$$\widetilde{\mathcal{L}}_{ab} = \delta_{ab}\left(\omega - \sum_{c:c\neq a}\frac{1}{t_a - t_c} - \sum_{k=1}^{n}\frac{1}{q_k - t_a}\right) + (1-\delta_{ab})\frac{1}{t_a - t_b}. \quad (M\times M)$$

Then $\det(\mathcal{L}) = \omega^{n-M}\det(\widetilde{\mathcal{L}})$.

Lemma

The matrices $\mathcal{L}, \widetilde{\mathcal{L}}$ can be represented as follows

$$\mathcal{L} = \omega + \mathcal{D} + D_0^{-1}(q)V^{T}(q)\,C_n^{T}\,V^{T}(q)^{-1}\,D_0(q)$$

$$\widetilde{\mathcal{L}} = \omega + \widetilde{\mathcal{D}} - D_0(t)\,V^{-1}(t)\,C_M\,V(t)\,D_0^{-1}(t)$$

Where

$$D_0(x_1,\dots,x_K) = \begin{pmatrix} \prod_{\ell\neq 1}(x_1 - x_\ell) & 0 & & 0 \\ 0 & \prod_{\ell\neq 2}(x_2 - x_\ell) & & \\ \vdots & 0 & \ddots & \\ 0 & \vdots & & \prod_{\ell\neq K}(x_K - x_\ell) \end{pmatrix} \left(= \delta_{ij}\prod_{k\neq i}(x_{ki} - x_\ell)\right).$$

$$V(x)_{ij} = x_j^{i-1}, \qquad \mathcal{D}_{ij} = \delta_{ij}\left(\sum_{a=1}^{M}\frac{1}{t_a - q_i}\right),$$

$$C_K = \sum_{j=1}^{1} j\cdot E_{j+1,j} = \begin{pmatrix} 0 & & & \\ 1 & 0 & & 0 \\ & 2 & \ddots & \\ 0 & & n-1 & 0 \end{pmatrix}. \qquad \widetilde{\mathcal{D}}_{ab} = \delta_{ab}\left(\sum_{k=1}^{n}\frac{1}{t_a - q_k}\right).$$

Proof

- How to invert Vandermonde matrix?

Claim $\left(V^{T}(x)\right)^{-1}_{ij} = (-1)^{n-j} e^{(i)}_{n-j} \cdot \frac{1}{\prod_{k,l\neq i}(x_i - x_l)} = \left(D_0^{-1}(x)\right)_{ii} \cdot \underbrace{(-1)^{n-j} e^{(i)}_{n-j}}_{\alpha_{ij}(x)}$

("exclude x_i")

Indeed: $\sum_{j=1}^{n} (-1)^{n-j} e^{(i)}_{n-j} x_k^{j-1} = \prod_{l\neq i}(x_k - x_l) = \begin{cases} 0, & k\neq i \\ \prod_{l\neq i}(x_i - x_l) \end{cases}$

$\Rightarrow V(x)^{-1} = D_0^{-1}(x)\,\alpha(x).$

Therefore

$$D_0(q)^{-1} V^{T}(q) C_n^{T} V^{T}(q)^{-1} D_0(q) = D_0(q)^{-1} V^{T}(q) C_n^{T} \alpha^{T}(q) \left(D_0^{-1}(q)\right)^{T} D_0(q)$$

$$\left(V^{T}(q) C_n^{T} \alpha^{T}(q)\right)_{ij} = \left(\alpha(q) C_n V(q)\right)_{ji} = \sum_{k=2}^{n} (-1)^{n-k} e^{(j)}_{n-k}(q)(k-1) q_i^{k-2} =$$

$$= \left(\prod_{k\neq j}(z - q_k)\right)'\Big|_{z=q_i}.$$

$$\Rightarrow \left(D_0(q)^{-1} V^{T}(q) C_n^{T} \alpha^{T}(q)\right)_{ij} = \frac{\left(\prod_{k\neq j}(z-q_k)\right)'\Big|_{z=q_i}}{\prod_{k\neq i}(q_i - q_k)} =$$

$$= \begin{cases} \dfrac{\prod_{k\neq j,i}(q_i - q_k)}{\prod_{k\neq i}(q_i - q_k)}, & i\neq j \\ \sum_{k\neq i} \dfrac{1}{q_i - q_k}, & i = j \end{cases} = \begin{cases} \dfrac{1}{q_i - q_j}, & i\neq j \\ \sum_{k\neq i} \dfrac{1}{q_i - q_k}, & i = j. \end{cases}$$

∎

Lemma

- $\det(\mathcal{L})$ has no poles at $q_i = q_j$, $(i \neq j)$
- $\det(\widetilde{\mathcal{L}})$ has no poles at $t_a = t_b$, $(a \neq b)$.

Proof.

$$\det(\mathcal{L}) = \det\left(\omega + \Phi + D_0^{-1}(q) V^T(q) C_n^T (V^T(q))^{-1} D_0(q)\right) =$$

$$= \det\left(\omega + \Phi + V^T(q) C_n^T (V^T(q))^{-1}\right) = \det\left(\omega + C_n^T + (V^T(q))^{-1} \Phi V^T(q)\right) =$$

$$= \det\left(\omega + C_n + V(q) \Phi (V(q))^{-1}\right).$$

$$\left(V(q)\Phi(V(q))^{-1}\right)_{ij} = \sum_{k=1}^{n} q_k^{i-1} \Phi_{kk} \cdot \frac{(-1)^{n-j} e_{n-j}^{(k)}}{D_0(q)_{kk}}$$

$$\sum_{j=1}^{n} p^{j-1} \left(V(q)\Phi V(q)^{-1}\right)_{ij} = \sum_{k=1}^{n} \frac{\Phi_{kk} q_k^{i-1}}{D_0(q)_{kk}} \prod_{\ell : \ell \neq k} (p - q_\ell) =$$

$$= \left(\prod_{\ell=1}^{n} (p - q_\ell)\right) \sum_{k=1}^{n} \frac{\Phi_{kk} q_k^{i-1}}{\left(\prod_{\ell \neq k} (q_k - q_\ell)\right)(p - q_k)}.$$

Residue at $q_r = q_s$ $[r < s)$:

$$\frac{\Phi_{rr} q_r^{i-1}}{\left(\prod_{\ell \neq r,s} (q_r - q_\ell)\right)(p - q_r)} - \frac{\Phi_{ss} q_s^{i-1}}{\left(\prod_{\ell \neq r,s} (q_s - q_\ell)\right)(p - q_s)} = 0.$$

($\Phi_{rr} = \Phi_{ss}$, $q_r = q_s$, $q_r = q_s$)

Proof for $\widetilde{\mathcal{L}}$ is similar. ∎

Proof of the Proposition:

Induction on $n+M$.

$$\operatorname*{Res}_{q_1=t_a}\left(\det(\mathcal{L})\right) = -\det\left(\mathcal{L}\left(\{q_i\}_{i>1}, \{t_b\}_{b\neq a}\right)\right) = -\omega^{(n-1)-(M-1)=n-M}\det\left(\mathcal{I}\left(\{q_i\}_{i>1}, \{t_b\}_{b\neq a}\right)\right)$$

$$\mathcal{L}_{ii\ (i>1)} \longrightarrow \omega + \sum_{\substack{k\neq i\\ k>1}} \frac{1}{q_i - q_k} + \cancel{\frac{1}{q_i - t_a}} + \cancel{\frac{1}{t_a - q_i}} + \sum_{\substack{b=1\\ b\neq a}}^{M} \frac{1}{t_b - q_i}$$

$$\operatorname*{Res}_{q_1=t_a}\left(\det(\widetilde{\mathcal{L}})\right) = -\det\left(\widetilde{\mathcal{L}}\left(\{q_i\}_{i>1}, \{t_b\}_{b\neq a}\right)\right)$$

$$\widetilde{\mathcal{L}}_{bb\ (b\neq a)} = \omega - \sum_{c:c\neq b} \frac{1}{t_b - t_c} - \cancel{\frac{1}{t_{ab} - t_a}} - \cancel{\frac{1}{t_a - t_b}} - \sum_{k>1} \frac{1}{q_k - t_{ab}}$$

As $q_1 \longrightarrow \infty$:

$$\det(\mathcal{L}) \longrightarrow \omega\cdot\det\left(\mathcal{L}\left(\{q_i\}_{i>1}, \{t_a\}\right)\right) = \omega\cdot\omega^{n-1-M}$$

$$\det(\widetilde{\mathcal{L}}) \longrightarrow \det\left(\widetilde{\mathcal{L}}\left(\{q_i\}_{i>1}, \{t_a\}\right)\right)$$

∎

Proof of the Theorem.

Introduce variables: $t_\beta^{(0)} = q_\beta \cdot \forall \beta = 1, \dots, n.$

$$t_1^{(0)}, \dots, t_n^{(0)} \,\Big|\, t_1^{(1)}, \dots, t_{N_1}^{(1)} \,\Big|\, \cdots \,\Big|\, t_1^{(\nu-1)}, \dots, t_{N_{\nu-1}}^{(\nu-1)} \,\Big|.$$

BAE:

$$v_{a+1} + \sum_{\beta':\beta'\neq\beta} \frac{1}{t_\beta^{(a)} - t_{\beta'}^{(a)}} + \sum_{\beta'} \frac{1}{t_{\beta'}^{(a+1)} - t_\beta^{(a)}} = v_a - \sum_{\beta':\beta'\neq\beta} \frac{1}{t_\beta^{(a)} - t_{\beta'}^{(a)}} -$$

$$- \sum_{\beta'} \frac{1}{t_{\beta'}^{(a-1)} - t_\beta^{(a)}}$$

Denote:

$$\mathcal{L}^{(a)}_{\alpha\beta} = \Big(v_{a+1} + \sum_{\beta':\beta'\neq\beta} \frac{1}{t^{(a)}_\beta - t^{(a)}_{\beta'}} + \sum_{\beta'} \frac{1}{t^{(a+1)}_{\beta'} - t^{(a)}_\beta} - \lambda\Big)\delta_{\alpha\beta} +$$

$$+ \frac{(1-\delta_{\alpha\beta})}{t^{(a)}_\alpha - t^{(a)}_\beta}.$$

Observe: • $\mathcal{L}^{(0)}_{\alpha\beta} = L_1^{CM}(\{h_\alpha\},\{q_\alpha\}) - \lambda$

• $\det(\mathcal{L}^{(a)}) = (v_{a+1}-\lambda)^{N_a - N_{a+1}} \cdot \det(\mathcal{L}^{(a+1)})$.

Indeed:

Apply proposition: $\omega \longrightarrow v_{a+1}-\lambda$

$\{q_i\} \longrightarrow \{t^{(a)}_\beta\}$

$\{t_a\} \longrightarrow \{t^{(a+1)}_\beta\}$

$$\det(\mathcal{L}^{(a)}) = (v_{a+1}-\lambda)^{N_a - N_{a+1}} \det(\tilde{\mathcal{L}}^{(a)}).$$

$$(\tilde{\mathcal{L}}^{(a)})_{\alpha\beta} = \Big(\underbrace{v_{a+1} - \lambda - \sum_{\beta':\beta'\neq\beta} \frac{1}{t^{(a+1)}_\beta - t^{(a+1)}_{\beta'}} - \sum_{\beta'} \frac{1}{t^{(a)}_{\beta'} - t^{(a+1)}_\beta}}_{\text{BAE } (a\to a+1)}\Big)\delta_{\alpha\beta} +$$

$$+ \frac{(1-\delta_{\alpha\beta})}{t^{(a+1)}_\alpha - t^{(a+1)}_\beta} =$$

$$v_{a+2} + \sum_{\beta':\beta'\neq\beta} \frac{1}{t^{(a+1)}_\beta - t^{(a+1)}_{\beta'}} + \sum_{\beta'} \Big(\frac{1}{t^{(a+2)}_{\beta'} - t^{(a+1)}_\beta}\Big) - \lambda$$

$$= \mathcal{L}^{(a+1)}_{\alpha\beta}$$

Hence $\det(L_1^{CM}(\{h_\alpha\},\{q_\alpha\}) - \lambda) = (v_1-\lambda)^{M-N_1}(v_2-\lambda)^{N_1-N_2}\cdot\ldots$

$$\cdot (v_N - \lambda)^{N_{N-1}-0} \cdot \underbrace{\det(\mathcal{L}^{(N+1)})}_{1}.$$

∎

Concluding remark

- This all can be deformed into correspondence between Ruijsenaars-Schneider model and XXX-chain.